

Real World Connection 1

Objective: My objective in this Virtual lab was to simulate how the process would be to build a simple network from scratch that could be used in a real world scenario, like a simple startup company with 1 network admin overseeing everything. For this lab I tried to stay as close as I can to a 3 core layer design. Each building has its own pair of VLANS that contain things such as printers and end hosts. I included a DHCP Server, NTP Server, DNS server and a Management Server in this network layout inside separate subnets to accurately depict workflow with different services throughout the network. I excluded the first 5 IP addresses of every VLAN to have a static IP for the SVI's, Printers, and Virtual IP addresses for HSRP. The layer 3 switches were responsible for SVI's/HSRP, ACLs, OSPF and IP Address Helper. As for the layer 2 switches, I used protocol PAgP to ensure redundant links improve the speed of traffic with trunk links being configured as well. Lastly, extended ACL rules were created to ensure workers can't access the servers that keep company production up and running.

Equipment: (14) 2911 Routers, (8) 2960 Switches, (29) PC's, (3) Servers, (9) 3560 Switches

Key Steps:

- a. Divide the network in separate buildings with 10.0.0.0/8 using VLSM to hold 4096, 2046, 1022, and 510 host with those being just for the workers
- b. Continue dividing it by /30 to make all the P2P connections
- c. Assign a VLAN to each subnet
- d. Create the DHCP server to ensure easy connection to the network while maintaining reserved IPs for SVI's and printers
- e. Configure PAgP on each switch with trunk links (assign the native VLAN to 330)
- f. Ensure that the PAgP connections between the 2960 and 3560 switches have 2 IP addresses from each VLAN with an HSRP configured to ensure failback safety
- g. Setup OSPF on each router/layer 3 switch so that it has full communication with every building
- h. Create a NTP server so that every router/switch has a matching timestamp to ensure easy troubleshooting for future incidents
- i. Create a DNS server to ensure easy connection to printers and also allows us to grab the IP address for remote-in assistance
- j. Create extended ACLs rules to ensure nobody in the workplace can access the servers besides the Management VLAN
- k. Create loopback interfaces on the routers/layer 3 switches to ensure we're not taking unnecessary bandwidth from the interfaces for easy SSH access
- l. Create a static Management VLAN with the address of 172.16.1.0/27 throughout the network to ensure we can configure network devices remotely

- m. Create access-class rules within every network device to ensure the PC in the management VLAN is the only one that has SSH access

Drawbacks :

The drawbacks for this network topology would be the connection between Router-E-1 and Router-E-2 being a single link instead of having multiple connections, after making the topology I should've had 3 E routers interconnecting with each other to have a failback safety. With that being said I only have 1 router within each building facing outwards and if one goes down then full communication is broken. Another issue I have with this topology is that my services like NTP, DHCP Server, and DNS are separate when in real world scenarios they are combined into 1 but I just really wanted to show network flow. One of the key configurations I missed in this topology was PAT due to having L3 switches that act as my default gateway and having my routers with different IPs and I couldn't implement double NAT. The other key configuration I missed was implementing security features like ARP Inspection, DHCP Snooping, and port-security due to the fact that etherchannel can't be combined together.

Real World Connection 1 (interfaces)

Switch-A-1	F0/1	Port-Channel 1	Multilayer-Switch-A-1
Switch-A-1	F0/2	Port-Channel 3	Switch-A-2
Switch-A-1	F0/3	Port-Channel 3	Switch-A-2
Switch-A-1	F0/4	Port-Channel 1	Multilayer-Switch-A-1
Switch-A-1	F0/5	Port-Channel 2	Multilayer-Switch-A-2
Switch-A-1	F0/6	Port-Channel 2	Multilayer-Switch-A-2
Switch-A-1	F0/7	PC 0	DHCP
Switch-A-1	G0/8	PC 2	DHCP
Switch-A-1	G0/9	PC 4	DHCP
Switch-A-1	F0/10	PC 20	DHCP
Switch-A-1	F0/11	PC 21	DHCP
Switch-A-1	F0/12	PC 22	DHCP
Switch-A-1	F0/13	Printer-A-10	10.0.0.4
Switch-A-1	F0/14	Printer-A-20	10.0.16.4
Switch-A-2	F0/1	Port-Channel 5	Multilayer-Switch-A-2
Switch-A-2	F0/2	Port-Channel 3	Switch-A-1
Switch-A-2	F0/3	Port-Channel 3	Switch-A-1
Switch-A-2	F0/4	Port-Channel 5	Multilayer-Switch-A-2
Switch-A-2	F0/5	Port-Channel 4	Multilayer-Switch-A-1
Switch-A-2	F0/6	Port-Channel 4	Multilayer-Switch-A-1
Switch-A-2	F0/7	PC 1	DHCP
Switch-A-2	F0/8	PC 3	DHCP
Switch-A-2	F0/9	PC 5	DHCP
Switch-A-2	F0/10	PC 23	DHCP
Switch-A-2	F0/11	PC 24	DHCP
Switch-A-2	F0/12	Printer-A-30	10.0.32.4
Multilayer-Switch-A-1	F0/1	Router-A-2	10.0.91.65
Multilayer-Switch-A-1	F0/2	Port-Channel 1	Switch-A-1
Multilayer-Switch-A-1	F0/3	Port-Channel 6	Multilayer-Switch-A-2
Multilayer-Switch-A-1	F0/4	Port-Channel 6	Multilayer-Switch-A-2

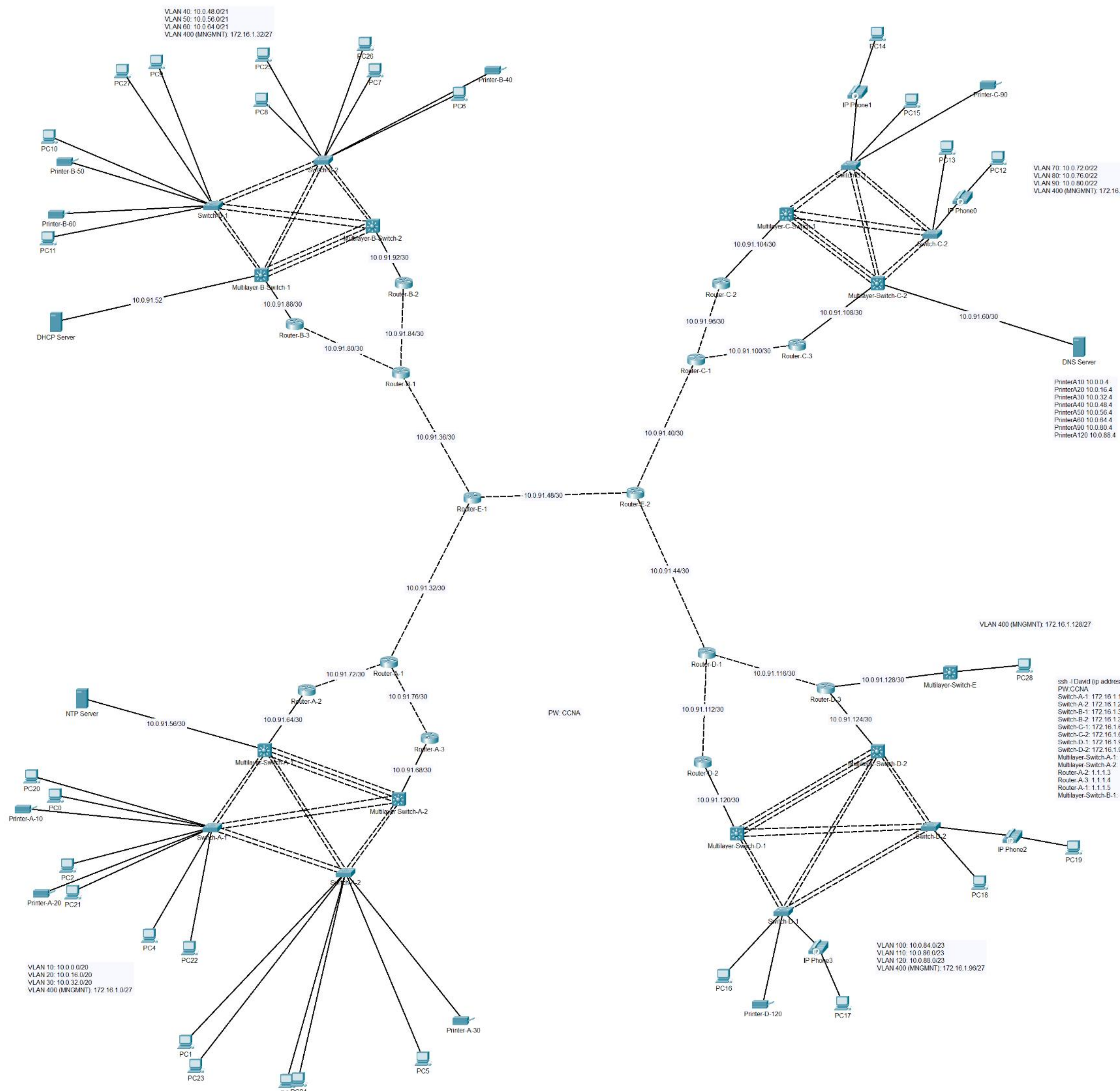
Multilayer-Switch-A-1	F0/5	Port-Channel 6	Multilayer-Switch-A-2
Multilayer-Switch-A-1	F0/6	Port-Channel 1	Switch-A-1
Multilayer-Switch-A-1	F0/7	NTP Server	10.0.91.57
Multilayer-Switch-A-1	F0/8	Port-Channel 4	Switch-A-2
Multilayer-Switch-A-1	F0/9	Port-Channel 4	Switch-A-2
Multilayer-Switch-A-2	F0/1	Router-A-3	10.0.91.69
Multilayer-Switch-A-2	F0/2	Port-Channel 5	Switch-A-2
Multilayer-Switch-A-2	F0/3	Port-Channel 6	Multilayer-Switch-A-1
Multilayer-Switch-A-2	F0/4	Port-Channel 6	Multilayer-Switch-A-1
Multilayer-Switch-A-2	F0/5	Port-Channel 6	Multilayer-Switch-A-1
Multilayer-Switch-A-2	F0/6	Port-Channel 5	Switch-A-2
Multilayer-Switch-A-2	F0/7	Port-Channel 2	Switch-A-1
Multilayer-Switch-A-2	F0/8	Port-Channel 2	Switch-A-1
Switch-B-1	F0/1	Port-Channel 5	Multilayer-Switch-B-1
Switch-B-1	F0/2	Port-Channel 3	Switch-B-2
Switch-B-1	F0/3	Port-Channel 3	Switch-B-2
Switch-B-1	F0/4	Port-Channel 4	Multilayer-Switch-B-2
Switch-B-1	F0/5	Port-Channel 4	Multilayer-Switch-B-2
Switch-B-1	F0/6	Port-Channel 5	Multilayer-Switch-B-1
Switch-B-1	F0/7	PC 9	DHCP
Switch-B-1	F0/8	PC 10	DHCP
Switch-B-1	F0/9	PC 11	DHCP
Switch-B-1	F0/10	PC 27	DHCP
Switch-B-1	F0/11	Printer-B-50	10.0.56.4
Switch-B-1	F0/12	Printer-B-60	10.0.64.4
Switch-B-2	F0/1	Port-Channel 1	Multilayer-Switch-B-2
Switch-B-2	F0/2	Port-Channel 3	Switch-B-1
Switch-B-2	F0/3	Port-Channel 3	Switch-B-1
Switch-B-2	F0/4	Port-Channel 1	Multilayer-Switch-B-2
Switch-B-2	F0/5	Port-Channel 2	Multilayer-Switch-B-1
Switch-B-2	F0/6	Port-Channel 2	Multilayer-Switch-B-1
Switch-B-2	F0/7	PC 6	DHCP
Switch-B-2	F0/8	PC 7	DHCP
Switch-B-2	F0/9	PC 8	DHCP
Switch-B-2	F0/10	Printer-B-40	10.0.48.4
Switch-B-2	F0/11	PC 26	DHCP
Switch-B-2	F0/12	PC 25	DHCP

Multilayer-Switch-B-1	F0/1	Router-B-3	10.0.91.89
Multilayer-Switch-B-1	F0/2	Port-Channel 5	Switch-B-1
Multilayer-Switch-B-1	F0/3	Port-Channel 6	Multilayer-Switch-B-2
Multilayer-Switch-B-1	F0/4	Port-Channel 6	Multilayer-Switch-B-2
Multilayer-Switch-B-1	F0/5	Port-Channel 6	Multilayer-Switch-B-2
Multilayer-Switch-B-1	F0/6	DHCP Server	10.0.91.53
Multilayer-Switch-B-1	F0/7	Port-Channel 5	Switch-B-1
Multilayer-Switch-B-1	F0/8	Port-Channel 2	Switch-B-2
Multilayer-Switch-B-1	F0/9	Port-Channel 2	Switch-B-2
Multilayer-Switch-B-2	F0/1	Router-B-2	10.0.91.93
Multilayer-Switch-B-2	F0/2	Port-Channel 1	Switch-B-2
Multilayer-Switch-B-2	F0/3	Port-Channel 6	Multilayer-Switch-B-1
Multilayer-Switch-B-2	F0/4	Port-Channel 6	Multilayer-Switch-B-1
Multilayer-Switch-B-2	F0/5	Port-Channel 6	Multilayer-Switch-B-1
Multilayer-Switch-B-2	F0/6	Port-Channel 4	Switch-B-1
Multilayer-Switch-B-2	F0/7	Port-Channel 4	Switch-B-1
Multilayer-Switch-B-2	F0/8	Port-Channel 1	Switch-B-2
Switch-C-1	F0/1	Port-Channel 5	Multilayer-Switch-C-1
Switch-C-1	F0/2	Port-Channel 5	Multilayer-Switch-C-1
Switch-C-1	F0/3	Port-Channel 3	Switch-C-2
Switch-C-1	F0/4	Port-Channel 3	Switch-C-2
Switch-C-1	F0/5	Port-Channel 4	Multilayer-Switch-C-2
Switch-C-1	F0/6	Port-Channel 4	Multilayer-Switch-C-2
Switch-C-1	F0/7	IP Phone/ PC 7	
Switch-C-1	F0/8	PC 15	DHCP
Switch-C-1	F0/9	Printer-C-90	10.0.80.4
Switch-C-2	F0/1	Port-Channel 1	Multilayer-Switch-C-2
Switch-C-2	F0/2	Port-Channel 1	Multilayer-Switch-C-2
Switch-C-2	F0/3	Port-Channel 3	Switch-C-1
Switch-C-2	F0/4	Port-Channel 3	Switch-C-1

Switch-C-2	F0/5	Port-Channel 2	Multilayer-Switch-C-1
Switch-C-2	F0/6	Port-Channel 2	Multilayer-Switch-C-1
Switch-C-2	F0/7	IP Phone/ PC 12	DHCP
Switch-C-2	F0/8	PC 13	DHCP
Multilayer-Switch-C-1	F0/1	Router-C-2	10.0.91.105
Multilayer-Switch-C-1	F0/2	Port-Channel 5	Switch-C-1
Multilayer-Switch-C-1	F0/3	Port-Channel 6	Multilayer-Switch-C-2
Multilayer-Switch-C-1	F0/4	Port-Channel 6	Multilayer-Switch-C-2
Multilayer-Switch-C-1	F0/5	Port-Channel 6	Multilayer-Switch-C-2
Multilayer-Switch-C-1	F0/6	Port-Channel 5	Switch-C-1
Multilayer-Switch-C-1	F0/7	Port-Channel 2	Switch-C-2
Multilayer-Switch-C-1	F0/8	Port-Channel 2	Switch-C-2
Multilayer-Switch-C-2	F0/1	Router-C-3	10.0.91.109
Multilayer-Switch-C-2	F0/2	Port-Channel 1	Switch-C-2
Multilayer-Switch-C-2	F0/3	Port-Channel 6	Multilayer-Switch-C-1
Multilayer-Switch-C-2	F0/4	Port-Channel 6	Multilayer-Switch-C-1
Multilayer-Switch-C-2	F0/5	Port-Channel 6	Multilayer-Switch-C-1
Multilayer-Switch-C-2	F0/6	Port-Channel 1	Switch-C-2
Multilayer-Switch-C-2	F0/7	DNS Server	10.0.91.61
Multilayer-Switch-C-2	F0/8	Port-Channel 4	Switch-C-1
Multilayer-Switch-C-2	F0/9	Port-Channel 4	Switch-C-1
Switch-D-1	F0/1	Port-Channel 1	Multilayer-Switch-D-1
Switch-D-1	F0/2	Port-Channel 3	Switch-D-2
Switch-D-1	F0/3	Port-Channel 3	Switch-D-2
Switch-D-1	F0/4	Port-Channel 2	Multilayer-Switch-D-2
Switch-D-1	F0/5	Port-Channel 2	Multilayer-Switch-D-2
Switch-D-1	F0/6	Port-Channel 1	Multilayer-Switch-D-1
Switch-D-1	F0/7	IP Phone / PC 17	DHCP
Switch-D-1	F0/8	PC 16	DHCP
Switch-D-1	F0/9	Printer-D-120	10.0.88.4
Switch-D-2	F0/1	Port-Channel 5	Multilayer-Switch-D-2
Switch-D-2	F0/2	Port-Channel 3	Switch-D-1
Switch-D-2	F0/3	Port-Channel 3	Switch-D-1
Switch-D-2	F0/4	Port-Channel 5	Multilayer-Switch-D-2
Switch-D-2	F0/5	Port-Channel 4	Multilayer-Switch-D-1
Switch-D-2	F0/6	Port-Channel 4	Multilayer-Switch-D-1
Switch-D-2	F0/7	IP Phone / PC 19	DHCP

Switch-D-2	F0/8	PC 18	DHCP
Multilayer-Switch-D-1	F0/1	Router-D-2	10.0.91.121
Multilayer-Switch-D-1	F0/2	Port-Channel 5	Switch-D-2
Multilayer-Switch-D-1	F0/3	Port-Channel 6	Multilayer-Switch-D-2
Multilayer-Switch-D-1	F0/4	Port-Channel 6	Multilayer-Switch-D-2
Multilayer-Switch-D-1	F0/5	Port-Channel 6	Multilayer-Switch-D-2
Multilayer-Switch-D-1	F0/6	Port-Channel 5	Switch-D-2
Multilayer-Switch-D-1	F0/7	Port-Channel 2	Switch-C-2
Multilayer-Switch-D-1	F0/8	Port-Channel 2	Switch-C-2
Multilayer-Switch-D-2	F0/1	Router-D-3	10.0.91.125
Multilayer-Switch-D-2	F0/2	Port-Channel 1	Switch-D-1
Multilayer-Switch-D-2	F0/3	Port-Channel 6	Multilayer-Switch-D-1
Multilayer-Switch-D-2	F0/4	Port-Channel 6	Multilayer-Switch-D-1
Multilayer-Switch-D-2	F0/5	Port-Channel 6	Multilayer-Switch-D-1
Multilayer-Switch-D-2	F0/6	Port-Channel 1	Switch-D-1
Multilayer-Switch-D-2	F0/7	Port-Channel 4	Switch-D-2
Multilayer-Switch-D-2	F0/8	Port-Channel 4	Switch-D-2
Router-A-1	G0/0	Router-E-1	10.0.91.33
Router-A-1	G0/1	Router-A-2	10.0.91.74
Router-A-1	G0/2	Router-A-3	10.0.91.78
Router-A-2	G0/0	Router-A-1	10.0.91.73
Router-A-2	G0/1	Multilayer-Switch-A-1	10.0.91.66
Router-A-3	G0/0	Router-A-1	10.0.91.77
Router-A-3	G0/1	Multilayer-Switch-A-2	10.0.91.70
Router-B-1	G0/0	Router-E-1	10.0.91.37
Router-B-1	G0/1	Router-B-2	10.0.91.86
Router-B-1	G0/2	Router-B-3	10.0.91.82
Router-B-2	G0/0	Router-B-1	10.0.91.85
Router-B-2	G0/1	Multilayer-Switch-B-2	10.0.91.94
Router-B-3	G0/0	Router-B-1	10.0.91.81
Router-B-3	G0/1	Multilayer-Switch-B-1	10.0.91.90
Router-C-1	G0/0	Router-E-2	10.0.91.41
Router-C-1	G0/1	Router-C-2	10.0.91.98
Router-C-1	G0/2	Router-C-3	10.0.91.102
Router-C-2	G0/0	Router-C-1	10.0.91.101
Router-C-2	G0/1	Multilayer-Switch-C-2	10.0.91.110

Router-C-3	G0/0	Router-C-1	10.0.91.97
Router-C-3	G0/1	Multilayer-Switch-C-1	10.0.91.106
Router-D-1	G0/0	Router-E-2	10.0.91.45
Router-D-1	G0/1	Router-D-3	10.0.91.118
Router-D-1	G0/2	Router-D-2	10.0.91.114
Router-D-2	G0/0	Router-D-1	10.0.91.113
Router-D-2	G0/1	Multilayer-Switch-D-1	10.0.91.122
Router-D-3	G0/0	Router-D-1	10.0.91.117
Router-D-3	G0/1	Multilayer-Switch-D-2	10.0.91.126
Router-D-3	G0/2	Multilayer-Switch-E	10.0.91.130
Router-E-1	G0/0	Router-E-2	10.0.91.49
Router-E-1	G0/1	Router-B-1	10.0.91.38
Router-E-1	G0/2	Router-A-1	10.0.91.34
Router-E-2	G0/0	Router-E-1	10.0.91.50
Router-E-2	G0/1	Router-C-1	10.0.48.42
Router-E-2	G0/2	Router-D-1	10.0.91.46



Switch-A-1:

```
hostname Switch-A-1
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip domain-name DavidsLab
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
no ip dhcp snooping information option
!
spanning-tree mode pvst
spanning-tree portfast default
spanning-tree portfast bpduguard default
spanning-tree extend system-id
spanning-tree vlan 30 priority 20480
spanning-tree vlan 400 priority 28672
!
interface Port-channel1
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport mode trunk
!
interface Port-channel2
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport mode trunk
!
interface Port-channel3
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport mode trunk
!
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/3
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/4
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport mode trunk
channel-group 2 mode desirable
```

```
!  
interface FastEthernet0/7  
  switchport access vlan 10  
  switchport mode access  
  switchport port-security  
  switchport port-security aging time 60  
!  
interface FastEthernet0/8  
  switchport access vlan 20  
  switchport mode access  
!  
interface FastEthernet0/9  
  switchport access vlan 30  
  switchport mode access  
  switchport port-security  
  switchport port-security aging time 60  
!  
interface FastEthernet0/10  
  switchport access vlan 10  
  switchport mode access  
!  
interface FastEthernet0/11  
  switchport access vlan 20  
  switchport mode access  
!  
interface FastEthernet0/12  
  switchport access vlan 30  
  switchport mode access  
!  
interface FastEthernet0/13  
  switchport access vlan 10  
  switchport mode access  
!  
interface FastEthernet0/14  
  switchport access vlan 20  
  switchport mode access  
!  
interface Vlan1  
  no ip address  
  shutdown  
!  
interface Vlan400  
  ip address 172.16.1.1 255.255.255.224  
!  
  ip default-gateway 172.16.1.30  
!  
!  
!  
!  
access-list 1 permit host 172.16.1.129  
line con 0  
!  
line vty 0 4  
  access-class 1 in  
  login local  
  transport input ssh  
line vty 5 15  
  access-class 1 in  
  login local  
  transport input ssh  
!  
!  
!  
!  
end
```

Switch-A-2:

Current configuration : 3454 bytes

```
!  
version 15.0  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname Switch-A-2  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
!  
!  
ip domain-name DavidsLab  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
!  
!  
spanning-tree mode pvst  
spanning-tree portfast default  
spanning-tree portfast bpduguard default  
spanning-tree extend system-id  
spanning-tree vlan 400 priority 24576  
spanning-tree vlan 30 priority 28672  
!  
interface Port-channel3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 10,20,30,400  
switchport mode trunk  
!  
interface Port-channel4  
switchport trunk native vlan 330  
switchport trunk allowed vlan 10,20,30,400  
switchport mode trunk  
!  
interface Port-channel5  
switchport trunk native vlan 330  
switchport trunk allowed vlan 10,20,30,400  
switchport mode trunk  
!  
interface FastEthernet0/1  
switchport trunk native vlan 330  
switchport trunk allowed vlan 10,20,30,400  
switchport mode trunk  
channel-group 5 mode desirable  
!  
interface FastEthernet0/2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 10,20,30,400  
switchport mode trunk  
channel-group 3 mode desirable  
!  
interface FastEthernet0/3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 10,20,30,400  
switchport mode trunk  
channel-group 3 mode desirable  
!  
interface FastEthernet0/4  
switchport trunk native vlan 330  
switchport trunk allowed vlan 10,20,30,400  
switchport mode trunk  
channel-group 5 mode desirable  
!  
interface FastEthernet0/5  
switchport trunk native vlan 330
```

```
switchport trunk allowed vlan 10,20,30,400
switchport mode trunk
channel-group 4 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport mode trunk
channel-group 4 mode desirable
!
interface FastEthernet0/7
switchport access vlan 10
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/8
switchport access vlan 20
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/9
switchport access vlan 30
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/10
switchport access vlan 10
switchport mode access
!
interface FastEthernet0/11
switchport access vlan 20
switchport mode access
!
interface FastEthernet0/12
switchport access vlan 30
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
interface Vlan400
ip address 172.16.1.2 255.255.255.224
!
ip default-gateway 172.16.1.30
!
!
!
!
access-list 1 permit host 172.16.1.129
line con 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
!
!
ntp server 10.0.91.58
!
end
```

Multilayer-Switch-1:

Current configuration : 6167 bytes

```
!  
version 12.2(37)SE1  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname Multilayer-Switch-A-1  
!  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip routing  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip domain-name DavidsLab  
!  
spanning-tree mode pvst  
spanning-tree vlan 10 priority 24576  
spanning-tree vlan 20 priority 28672  
!  
interface Loopback0  
ip address 1.1.1.1 255.255.255.255  
!  
interface Port-channel1  
switchport trunk native vlan 330  
switchport trunk allowed vlan 10,20,30,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface Port-channel4  
switchport trunk native vlan 330  
switchport trunk allowed vlan 10,20,30,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface Port-channel6  
switchport trunk native vlan 330  
switchport trunk allowed vlan 10,20,30,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface FastEthernet0/1  
no switchport  
ip address 10.0.91.65 255.255.255.252  
duplex auto  
speed auto  
!  
interface FastEthernet0/2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 10,20,30,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
channel-group 1 mode desirable  
!  
interface FastEthernet0/3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 10,20,30,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
channel-group 6 mode desirable  
!  
interface FastEthernet0/4  
switchport trunk native vlan 330
```

```
switchport trunk allowed vlan 10,20,30,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/7
no switchport
ip address 10.0.91.57 255.255.255.252
duplex auto
speed auto
!
interface FastEthernet0/8
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 4 mode desirable
!
interface FastEthernet0/9
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 4 mode desirable
!
interface Vlan1
no ip address
shutdown
!
interface Vlan10
mac-address 00d0.5837.ce01
ip address 10.0.0.2 255.255.240.0
ip helper-address 10.0.91.54
ip access-group To_Network10 in
standby version 2
standby 0 ip 10.0.0.1
standby 0 priority 105
standby 0 preempt
!
interface Vlan20
mac-address 00d0.5837.ce02
ip address 10.0.16.2 255.255.240.0
ip helper-address 10.0.91.54
ip access-group To_Network20 in
standby version 2
standby 2 ip 10.0.16.1
standby 0 preempt
!
interface Vlan30
mac-address 00d0.5837.ce03
ip address 10.0.32.2 255.255.240.0
ip helper-address 10.0.91.54
ip access-group To_Network30 in
standby version 2
standby 3 ip 10.0.32.1
standby 0 preempt
```

```

!
interface Vlan400
 mac-address 00d0.5837.ce04
 ip address 172.16.1.28 255.255.255.224
 standby version 2
 standby 400 ip 172.16.1.30
 standby 0 preempt
!
router ospf 1
 log-adjacency-changes
 network 10.0.91.64 0.0.0.3 area 0
 network 10.0.91.56 0.0.0.3 area 0
 network 10.0.0.0 0.0.15.255 area 0
 network 10.0.16.0 0.0.15.255 area 0
 network 10.0.32.0 0.0.15.255 area 0
 network 172.16.1.0 0.0.0.31 area 0
 network 1.1.1.1 0.0.0.0 area 0
!
ip classless
!
ip flow-export version 9
!
ip access-list extended To_Network10
 permit udp 10.0.0.0 0.0.15.255 host 10.0.91.58 eq 123
 permit udp 10.0.0.0 0.0.15.255 host 10.0.91.54 eq bootps
 permit udp 10.0.0.0 0.0.15.255 host 10.0.91.54 eq bootpc
 permit tcp 10.0.0.0 0.0.15.255 host 10.0.91.62 eq domain
 permit udp 10.0.0.0 0.0.15.255 host 10.0.91.62 eq domain
 deny ip 10.0.0.0 0.0.15.255 172.16.1.0 0.0.0.255
 deny ip 10.0.0.0 0.0.15.255 10.0.91.56 0.0.0.3
 deny ip 10.0.0.0 0.0.15.255 10.0.91.52 0.0.0.3
 deny ip 10.0.0.0 0.0.15.255 10.0.91.60 0.0.0.3
 permit ip any any
ip access-list extended To_Network20
 permit udp 10.0.16.0 0.0.15.255 host 10.0.91.58 eq 123
 permit udp 10.0.16.0 0.0.15.255 host 10.0.91.54 eq bootps
 permit udp 10.0.16.0 0.0.15.255 host 10.0.91.54 eq bootpc
 permit tcp 10.0.16.0 0.0.15.255 host 10.0.91.62 eq domain
 permit udp 10.0.16.0 0.0.15.255 host 10.0.91.62 eq domain
 deny ip 10.0.16.0 0.0.15.255 172.16.1.0 0.0.0.255
 deny ip 10.0.16.0 0.0.15.255 10.0.91.56 0.0.0.3
 deny ip 10.0.16.0 0.0.15.255 10.0.91.52 0.0.0.3
 deny ip 10.0.16.0 0.0.15.255 10.0.91.60 0.0.0.3
 permit ip any any
ip access-list extended To_Network30
 permit udp 10.0.32.0 0.0.15.255 host 10.0.91.58 eq 123
 permit udp 10.0.32.0 0.0.15.255 host 10.0.91.54 eq bootps
 permit udp 10.0.32.0 0.0.15.255 host 10.0.91.54 eq bootpc
 permit tcp 10.0.32.0 0.0.15.255 host 10.0.91.62 eq domain
 permit udp 10.0.32.0 0.0.15.255 host 10.0.91.62 eq domain
 deny ip 10.0.32.0 0.0.15.255 172.16.1.0 0.0.0.255
 deny ip 10.0.32.0 0.0.15.255 10.0.91.56 0.0.0.3
 deny ip 10.0.32.0 0.0.15.255 10.0.91.52 0.0.0.3
 deny ip 10.0.32.0 0.0.15.255 10.0.91.60 0.0.0.3
 permit ip any any
access-list 1 permit host 172.16.1.129
!
line con 0
!
line aux 0
!
line vty 0 4
 access-class 1 in
 login local
 transport input ssh
line vty 5 15
 access-class 1 in
 login local
 transport input ssh
!

```



```
ntp server 10.0.91.58
!
end
```

Multilayer-Switch-2:

```
!
hostname Multilayer-Switch-A-2
!
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip routing
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip domain-name DavidsLab
!
spanning-tree mode pvst
spanning-tree vlan 20 priority 24576
spanning-tree vlan 10 priority 28672
!
interface Loopback0
ip address 1.1.1.2 255.255.255.255
!
interface Port-channel2
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Port-channel5
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Port-channel6
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface FastEthernet0/1
no switchport
ip address 10.0.91.69 255.255.255.252
duplex auto
speed auto
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 5 mode desirable
!
interface FastEthernet0/3
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/4
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
```

```

!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 5 mode desirable
!
interface FastEthernet0/7
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/8
switchport trunk native vlan 330
switchport trunk allowed vlan 10,20,30,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 2 mode desirable
!
interface Vlan1
no ip address
shutdown
!
interface Vlan10
mac-address 0090.2b1b.9801
ip address 10.0.0.3 255.255.240.0
ip helper-address 10.0.91.54
ip access-group To_Network10 in
standby version 2
standby 0 ip 10.0.0.1
standby 0 priority 90
!
interface Vlan20
mac-address 0090.2b1b.9802
ip address 10.0.16.3 255.255.240.0
ip helper-address 10.0.91.54
ip access-group To_Network20 in
standby version 2
standby 2 ip 10.0.16.1
!
interface Vlan30
mac-address 0090.2b1b.9803
ip address 10.0.32.3 255.255.240.0
ip helper-address 10.0.91.54
ip access-group To_Network30 in
standby version 2
standby 3 ip 10.0.32.1
!
interface Vlan400
mac-address 0090.2b1b.9804
ip address 172.16.1.29 255.255.255.224
standby version 2
standby 400 ip 172.16.1.30
!
router ospf 1
log-adjacency-changes
network 10.0.91.68 0.0.0.3 area 0
network 10.0.0.0 0.0.15.255 area 0
network 10.0.16.0 0.0.15.255 area 0
network 10.0.32.0 0.0.15.255 area 0

```

```

network 172.16.1.0 0.0.0.31 area 0
network 1.1.1.2 0.0.0.0 area 0
!
ip classless
!
ip flow-export version 9
!
!
ip access-list extended To_Network10
permit udp 10.0.0.0 0.0.15.255 host 10.0.91.58 eq 123
permit udp 10.0.0.0 0.0.15.255 host 10.0.91.54 eq bootps
permit udp 10.0.0.0 0.0.15.255 host 10.0.91.54 eq bootpc
permit tcp 10.0.0.0 0.0.15.255 host 10.0.91.62 eq domain
permit udp 10.0.0.0 0.0.15.255 host 10.0.91.62 eq domain
deny ip 10.0.0.0 0.0.15.255 172.16.1.0 0.0.0.255
deny ip 10.0.0.0 0.0.15.255 10.0.91.56 0.0.0.3
deny ip 10.0.0.0 0.0.15.255 10.0.91.52 0.0.0.3
deny ip 10.0.0.0 0.0.15.255 10.0.91.60 0.0.0.3
permit ip any any
ip access-list extended To_Network20
permit udp 10.0.16.0 0.0.15.255 host 10.0.91.58 eq 123
permit udp 10.0.16.0 0.0.15.255 host 10.0.91.54 eq bootps
permit udp 10.0.16.0 0.0.15.255 host 10.0.91.54 eq bootpc
permit tcp 10.0.16.0 0.0.15.255 host 10.0.91.62 eq domain
permit udp 10.0.16.0 0.0.15.255 host 10.0.91.62 eq domain
deny ip 10.0.16.0 0.0.15.255 172.16.1.0 0.0.0.255
deny ip 10.0.16.0 0.0.15.255 10.0.91.56 0.0.0.3
deny ip 10.0.16.0 0.0.15.255 10.0.91.52 0.0.0.3
deny ip 10.0.16.0 0.0.15.255 10.0.91.60 0.0.0.3
permit ip any any
ip access-list extended To_Network30
permit udp 10.0.32.0 0.0.15.255 host 10.0.91.58 eq 123
permit udp 10.0.32.0 0.0.15.255 host 10.0.91.54 eq bootps
permit udp 10.0.32.0 0.0.15.255 host 10.0.91.54 eq bootpc
permit tcp 10.0.32.0 0.0.15.255 host 10.0.91.62 eq domain
permit udp 10.0.32.0 0.0.15.255 host 10.0.91.62 eq domain
deny ip 10.0.32.0 0.0.15.255 172.16.1.0 0.0.0.255
deny ip 10.0.32.0 0.0.15.255 10.0.91.56 0.0.0.3
deny ip 10.0.32.0 0.0.15.255 10.0.91.52 0.0.0.3
deny ip 10.0.32.0 0.0.15.255 10.0.91.60 0.0.0.3
permit ip any any
access-list 1 permit host 172.16.1.129
!
line con 0
!
line aux 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
ntp server 10.0.91.58
!
end

```

Switch-B-1:

Building configuration...

Current configuration : 3607 bytes

```
!  
version 15.0  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname Switch-B-1  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip domain-name DavidsLab  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
spanning-tree mode pvst  
spanning-tree portfast default  
spanning-tree portfast bpduguard default  
spanning-tree extend system-id  
spanning-tree vlan 60 priority 20480  
spanning-tree vlan 400 priority 28672  
!  
interface Port-channel3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport mode trunk  
!  
interface Port-channel4  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport mode trunk  
!  
interface Port-channel5  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport mode trunk  
!  
interface FastEthernet0/1  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport mode trunk  
channel-group 5 mode desirable  
!  
interface FastEthernet0/2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport mode trunk  
channel-group 3 mode desirable  
!  
interface FastEthernet0/3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport mode trunk  
channel-group 3 mode desirable  
!  
interface FastEthernet0/4  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport mode trunk
```

```
channel-group 4 mode desirable
!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,400
switchport mode trunk
channel-group 4 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,400
switchport mode trunk
channel-group 5 mode desirable
!
interface FastEthernet0/7
switchport access vlan 40
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/8
switchport access vlan 50
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/9
switchport access vlan 60
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/10
switchport access vlan 40
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/11
switchport access vlan 50
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/12
switchport access vlan 60
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface Vlan1
no ip address
shutdown
!
interface Vlan400
ip address 172.16.1.33 255.255.255.224
!
ip default-gateway 172.16.1.62
!
access-list 1 permit host 172.16.1.129
line con 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
```

```
!  
end
```

Switch-B-2:

```
!  
hostname Switch-B-2  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip domain-name DavidsLab  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
spanning-tree mode pvst  
spanning-tree portfast default  
spanning-tree portfast bpduguard default  
spanning-tree extend system-id  
spanning-tree vlan 400 priority 24576  
spanning-tree vlan 60 priority 28672  
!  
interface Port-channel1  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport mode trunk  
!  
interface Port-channel2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport mode trunk  
!  
interface Port-channel3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport mode trunk  
!  
interface FastEthernet0/1  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport mode trunk  
channel-group 1 mode desirable  
!  
interface FastEthernet0/2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport mode trunk  
channel-group 3 mode desirable  
!  
interface FastEthernet0/3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport mode trunk  
channel-group 3 mode desirable  
!  
interface FastEthernet0/4  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport mode trunk  
channel-group 1 mode desirable  
!  
interface FastEthernet0/5  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport mode trunk
```

```
channel-group 2 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,400
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/7
switchport access vlan 40
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/8
switchport access vlan 50
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/9
switchport access vlan 60
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/10
switchport access vlan 40
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/11
switchport access vlan 50
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface FastEthernet0/12
switchport access vlan 60
switchport mode access
switchport port-security
switchport port-security aging time 60
!
interface Vlan1
no ip address
shutdown
!
interface Vlan400
ip address 172.16.1.34 255.255.255.224
!
ip default-gateway 172.16.1.62
!
!
!
!
access-list 1 permit host 172.16.1.129
line con 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
!
!
```

```
!  
end
```

Multilayer-B-Switch-1:

```
!  
hostname Multilayer-Switch-B-1  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip routing  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip domain-name DavidsLab  
!  
spanning-tree mode pvst  
spanning-tree vlan 50 priority 24576  
spanning-tree vlan 40 priority 28672  
!  
interface Loopback0  
ip address 1.1.1.6 255.255.255.255  
!  
interface Port-channel2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface Port-channel5  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface Port-channel6  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface FastEthernet0/1  
no switchport  
ip address 10.0.91.89 255.255.255.252  
duplex auto  
speed auto  
!  
interface FastEthernet0/2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
channel-group 5 mode desirable  
!  
interface FastEthernet0/3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
channel-group 6 mode desirable  
!  
interface FastEthernet0/4
```



```
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/6
no switchport
ip address 10.0.91.53 255.255.255.252
duplex auto
speed auto
!
interface FastEthernet0/7
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 5 mode desirable
!
interface FastEthernet0/8
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/9
switchport trunk native vlan 330
switchport trunk allowed vlan 40,50,60,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 2 mode desirable
!
interface Vlan1
no ip address
shutdown
!
interface Vlan40
mac-address 0001.96e0.6a01
ip address 10.0.48.3 255.255.248.0
ip helper-address 10.0.91.54
ip access-group To_Network40 in
standby version 2
standby 4 ip 10.0.48.1
standby 0 preempt
!
interface Vlan50
mac-address 0001.96e0.6a02
ip address 10.0.56.3 255.255.248.0
ip helper-address 10.0.91.54
ip access-group To_Network50 in
standby version 2
standby 5 ip 10.0.56.1
!
interface Vlan60
mac-address 0001.96e0.6a03
ip address 10.0.64.3 255.255.248.0
ip helper-address 10.0.91.54
ip access-group To_Network60 in
standby version 2
standby 6 ip 10.0.64.1
!
interface Vlan400
```

```

mac-address 0001.96e0.6a04
ip address 172.16.1.60 255.255.255.224
standby version 2
standby 400 ip 172.16.1.62
standby 0 preempt
!
router ospf 1
log-adjacency-changes
network 10.0.91.88 0.0.0.3 area 0
network 10.0.91.52 0.0.0.3 area 0
network 10.0.48.0 0.0.7.255 area 0
network 10.0.56.0 0.0.7.255 area 0
network 10.0.64.0 0.0.7.255 area 0
network 172.16.1.32 0.0.0.31 area 0
network 1.1.1.6 0.0.0.0 area 0
!
ip classless
!
ip flow-export version 9
!
ip access-list extended To_Network40
permit udp 10.0.48.0 0.0.7.255 host 10.0.91.58 eq 123
permit udp 10.0.48.0 0.0.7.255 host 10.0.91.54 eq bootps
permit udp 10.0.48.0 0.0.7.255 host 10.0.91.54 eq bootpc
permit udp 10.0.48.0 0.0.7.255 host 10.0.91.62 eq domain
permit tcp 10.0.48.0 0.0.7.255 host 10.0.91.62 eq domain
deny ip 10.0.48.0 0.0.7.255 172.16.1.0 0.0.0.255
deny ip 10.0.48.0 0.0.7.255 10.0.91.56 0.0.0.3
deny ip 10.0.48.0 0.0.7.255 10.0.91.52 0.0.0.3
deny ip 10.0.48.0 0.0.7.255 10.0.91.60 0.0.0.3
permit ip any any
ip access-list extended To_Network50
permit udp 10.0.56.0 0.0.7.255 host 10.0.91.58 eq 123
permit udp 10.0.56.0 0.0.7.255 host 10.0.91.54 eq bootps
permit udp 10.0.56.0 0.0.7.255 host 10.0.91.54 eq bootpc
permit udp 10.0.56.0 0.0.7.255 host 10.0.91.62 eq domain
permit tcp 10.0.56.0 0.0.7.255 host 10.0.91.62 eq domain
deny ip 10.0.56.0 0.0.7.255 172.16.1.0 0.0.0.255
deny ip 10.0.56.0 0.0.7.255 10.0.91.56 0.0.0.3
deny ip 10.0.56.0 0.0.7.255 10.0.91.52 0.0.0.3
deny ip 10.0.56.0 0.0.7.255 10.0.91.60 0.0.0.3
permit ip any any
ip access-list extended To_Network60
permit udp 10.0.64.0 0.0.7.255 host 10.0.91.58 eq 123
permit udp 10.0.64.0 0.0.7.255 host 10.0.91.54 eq bootps
permit udp 10.0.64.0 0.0.7.255 host 10.0.91.54 eq bootpc
permit udp 10.0.64.0 0.0.7.255 host 10.0.91.62 eq domain
permit tcp 10.0.64.0 0.0.7.255 host 10.0.91.62 eq domain
deny ip 10.0.64.0 0.0.7.255 172.16.1.0 0.0.0.255
deny ip 10.0.64.0 0.0.7.255 10.0.91.56 0.0.0.3
deny ip 10.0.64.0 0.0.7.255 10.0.91.52 0.0.0.3
deny ip 10.0.64.0 0.0.7.255 10.0.91.60 0.0.0.3
permit ip any any
access-list 1 permit host 172.16.1.129
!
line con 0
!
line aux 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
ntp server 10.0.91.58
!

```

end

Multilayer-B-Switch-2:

```
!  
hostname Multilayer-Switch-B-2  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip routing  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip domain-name DavidsLab  
!  
spanning-tree mode pvst  
spanning-tree vlan 40 priority 24576  
spanning-tree vlan 50 priority 28672  
!  
interface Loopback0  
ip address 1.1.1.7 255.255.255.255  
!  
interface Port-channel1  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface Port-channel4  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface Port-channel6  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface FastEthernet0/1  
no switchport  
ip address 10.0.91.93 255.255.255.252  
duplex auto  
speed auto  
!  
interface FastEthernet0/2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
channel-group 1 mode desirable  
!  
interface FastEthernet0/3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 40,50,60,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
channel-group 6 mode desirable
```

```
!  
interface FastEthernet0/4  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 40,50,60,400  
  switchport trunk encapsulation dot1q  
  switchport mode trunk  
  channel-group 6 mode desirable  
!  
interface FastEthernet0/5  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 40,50,60,400  
  switchport trunk encapsulation dot1q  
  switchport mode trunk  
  channel-group 6 mode desirable  
!  
interface FastEthernet0/6  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 40,50,60,400  
  switchport trunk encapsulation dot1q  
  switchport mode trunk  
  channel-group 4 mode desirable  
!  
interface FastEthernet0/7  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 40,50,60,400  
  switchport trunk encapsulation dot1q  
  switchport mode trunk  
  channel-group 4 mode desirable  
!  
interface FastEthernet0/8  
  switchport trunk native vlan 330  
  switchport trunk allowed vlan 40,50,60,400  
  switchport trunk encapsulation dot1q  
  switchport mode trunk  
  channel-group 1 mode desirable  
!  
interface Vlan1  
  no ip address  
  shutdown  
!  
interface Vlan40  
  mac-address 0060.3ee4.a601  
  ip address 10.0.48.2 255.255.248.0  
  ip helper-address 10.0.91.54  
  ip access-group To_Network40 in  
  standby version 2  
  standby 4 ip 10.0.48.1  
!  
interface Vlan50  
  mac-address 0060.3ee4.a602  
  ip address 10.0.56.2 255.255.248.0  
  ip helper-address 10.0.91.54  
  ip access-group To_Network50 in  
  standby version 2  
  standby 5 ip 10.0.56.1  
  standby 0 preempt  
!  
interface Vlan60  
  mac-address 0060.3ee4.a603  
  ip address 10.0.64.2 255.255.248.0  
  ip helper-address 10.0.91.54  
  ip access-group To_Network60 in  
  standby version 2  
  standby 6 ip 10.0.64.1  
  standby 0 preempt  
!  
interface Vlan400  
  mac-address 0060.3ee4.a604  
  ip address 172.16.1.61 255.255.255.224  
  standby version 2
```

```

standby 400 ip 172.16.1.62
!
router ospf 1
log-adjacency-changes
network 10.0.91.92 0.0.0.3 area 0
network 10.0.48.0 0.0.7.255 area 0
network 10.0.56.0 0.0.7.255 area 0
network 10.0.64.0 0.0.7.255 area 0
network 172.16.1.32 0.0.0.31 area 0
network 1.1.1.7 0.0.0.0 area 0
!
ip classless
!
ip flow-export version 9
!
ip access-list extended To_Network40
permit udp 10.0.48.0 0.0.7.255 host 10.0.91.58 eq 123
permit udp 10.0.48.0 0.0.7.255 host 10.0.91.54 eq bootps
permit udp 10.0.48.0 0.0.7.255 host 10.0.91.54 eq bootpc
permit udp 10.0.48.0 0.0.7.255 host 10.0.91.62 eq domain
permit tcp 10.0.48.0 0.0.7.255 host 10.0.91.62 eq domain
deny ip 10.0.48.0 0.0.7.255 172.16.1.0 0.0.0.255
deny ip 10.0.48.0 0.0.7.255 10.0.91.56 0.0.0.3
deny ip 10.0.48.0 0.0.7.255 10.0.91.52 0.0.0.3
deny ip 10.0.48.0 0.0.7.255 10.0.91.60 0.0.0.3
permit ip any any
ip access-list extended To_Network50
permit udp 10.0.56.0 0.0.7.255 host 10.0.91.58 eq 123
permit udp 10.0.56.0 0.0.7.255 host 10.0.91.54 eq bootps
permit udp 10.0.56.0 0.0.7.255 host 10.0.91.54 eq bootpc
permit udp 10.0.56.0 0.0.7.255 host 10.0.91.62 eq domain
permit tcp 10.0.56.0 0.0.7.255 host 10.0.91.62 eq domain
deny ip 10.0.56.0 0.0.7.255 172.16.1.0 0.0.0.255
deny ip 10.0.56.0 0.0.7.255 10.0.91.56 0.0.0.3
deny ip 10.0.56.0 0.0.7.255 10.0.91.52 0.0.0.3
deny ip 10.0.56.0 0.0.7.255 10.0.91.60 0.0.0.3
permit ip any any
ip access-list extended To_Network60
permit udp 10.0.64.0 0.0.7.255 host 10.0.91.58 eq 123
permit udp 10.0.64.0 0.0.7.255 host 10.0.91.54 eq bootps
permit udp 10.0.64.0 0.0.7.255 host 10.0.91.54 eq bootpc
permit udp 10.0.64.0 0.0.7.255 host 10.0.91.62 eq domain
permit tcp 10.0.64.0 0.0.7.255 host 10.0.91.62 eq domain
deny ip 10.0.64.0 0.0.7.255 172.16.1.0 0.0.0.255
deny ip 10.0.64.0 0.0.7.255 10.0.91.56 0.0.0.3
deny ip 10.0.64.0 0.0.7.255 10.0.91.52 0.0.0.3
deny ip 10.0.64.0 0.0.7.255 10.0.91.60 0.0.0.3
permit ip any any
access-list 1 permit host 172.16.1.129
!
line con 0
!
line aux 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
ntp server 10.0.91.58
!
end

```

Switch-C-1:

```
!  
hostname Switch-C-1  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip domain-name DavidsLab  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
spanning-tree mode pvst  
spanning-tree portfast default  
spanning-tree portfast bpduguard default  
spanning-tree extend system-id  
spanning-tree vlan 90 priority 24576  
spanning-tree vlan 400 priority 28672  
!  
interface Port-channel3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90,400  
switchport mode trunk  
!  
interface Port-channel4  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90,400  
switchport mode trunk  
!  
interface Port-channel5  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90,400  
switchport mode trunk  
!  
interface FastEthernet0/1  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90,400  
switchport mode trunk  
channel-group 5 mode desirable  
!  
interface FastEthernet0/2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90,400  
switchport mode trunk  
channel-group 5 mode desirable  
!  
interface FastEthernet0/3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90,400  
switchport mode trunk  
channel-group 3 mode desirable  
!  
interface FastEthernet0/4  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90,400  
switchport mode trunk
```

```

channel-group 3 mode desirable
!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport mode trunk
channel-group 4 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport mode trunk
channel-group 4 mode desirable
!
interface FastEthernet0/7
switchport access vlan 80
switchport mode access
switchport voice vlan 70
!
interface FastEthernet0/8
switchport access vlan 90
switchport mode access
!
interface FastEthernet0/9
switchport access vlan 90
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
interface Vlan400
ip address 172.16.1.65 255.255.255.224
!
ip default-gateway 172.16.1.94
!
!
!
!
access-list 1 permit host 172.16.1.129
line con 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
!
!
!
end

```

Switch-C-2:

```

!
hostname Switch-C-2
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip domain-name DavidsLab
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
spanning-tree mode pvst

```

```
spanning-tree portfast default
spanning-tree portfast bpduguard default
spanning-tree extend system-id
spanning-tree vlan 400 priority 24576
spanning-tree vlan 90 priority 28672
!
interface Port-channel1
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport mode trunk
!
interface Port-channel2
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport mode trunk
!
interface Port-channel3
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport mode trunk
!
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/3
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/4
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/7
switchport access vlan 80
switchport mode access
switchport voice vlan 70
!
interface FastEthernet0/8
switchport access vlan 90
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
```



```

interface Vlan400
ip address 172.16.1.66 255.255.255.224
!
ip default-gateway 172.16.1.94
!
access-list 1 permit host 172.16.1.129
line con 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
end

```

Multilayer-Switch-C-1:

```

!
hostname Multilayer-Switch-C-1
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip routing
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip domain-name DavidsLab
!
spanning-tree mode pvst
spanning-tree vlan 80 priority 24576
spanning-tree vlan 70 priority 28672
!
interface Loopback0
ip address 1.1.1.11 255.255.255.255
!
interface Port-channel2
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Port-channel5
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Port-channel6
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface FastEthernet0/1
no switchport
ip address 10.0.91.106 255.255.255.252
duplex auto
speed auto
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 5 mode desirable

```

```

!
interface FastEthernet0/3
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/4
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 5 mode desirable
!
interface FastEthernet0/7
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/8
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 2 mode desirable
!
interface Vlan1
no ip address
shutdown
!
interface Vlan70
mac-address 00e0.b05e.2a01
ip address 10.0.72.3 255.255.252.0
ip helper-address 10.0.91.54
ip access-group To_Network70 in
standby version 2
standby 7 ip 10.0.72.1
!
interface Vlan80
mac-address 00e0.b05e.2a02
ip address 10.0.76.3 255.255.252.0
ip helper-address 10.0.91.54
ip access-group To_Network80 in
standby version 2
standby 8 ip 10.0.76.1
!
interface Vlan90
mac-address 00e0.b05e.2a03
ip address 10.0.80.3 255.255.252.0
ip helper-address 10.0.91.54
ip access-group To_Network90 in
standby version 2
standby 9 ip 10.0.80.1

```

```

!
interface Vlan400
  mac-address 00e0.b05e.2a04
  ip address 172.16.1.92 255.255.255.224
  standby version 2
  standby 400 ip 172.16.1.94
  standby 0 preempt
!
router ospf 1
  log-adjacency-changes
  network 10.0.91.104 0.0.0.3 area 0
  network 10.0.72.0 0.0.3.255 area 0
  network 10.0.76.0 0.0.3.255 area 0
  network 10.0.80.0 0.0.3.255 area 0
  network 172.16.1.64 0.0.0.31 area 0
  network 1.1.1.11 0.0.0.0 area 0
!
ip classless
!
ip flow-export version 9
!
ip access-list extended To_Network70
  permit udp 10.0.72.0 0.0.3.255 host 10.0.91.58 eq 123
  permit udp 10.0.72.0 0.0.3.255 host 10.0.91.54 eq bootps
  permit udp 10.0.72.0 0.0.3.255 host 10.0.91.54 eq bootpc
  permit udp 10.0.72.0 0.0.3.255 host 10.0.91.62 eq domain
  permit tcp 10.0.72.0 0.0.3.255 host 10.0.91.62 eq domain
  deny ip 10.0.72.0 0.0.3.255 172.16.1.0 0.0.0.255
  deny ip 10.0.72.0 0.0.3.255 10.0.91.56 0.0.0.3
  deny ip 10.0.72.0 0.0.3.255 10.0.91.52 0.0.0.3
  deny ip 10.0.72.0 0.0.3.255 10.0.91.60 0.0.0.3
  permit ip any any
ip access-list extended To_Network80
  permit udp 10.0.76.0 0.0.3.255 host 10.0.91.58 eq 123
  permit udp 10.0.76.0 0.0.3.255 host 10.0.91.54 eq bootps
  permit udp 10.0.76.0 0.0.3.255 host 10.0.91.54 eq bootpc
  permit udp 10.0.76.0 0.0.3.255 host 10.0.91.62 eq domain
  permit tcp 10.0.76.0 0.0.3.255 host 10.0.91.62 eq domain
  deny ip 10.0.76.0 0.0.3.255 172.16.1.0 0.0.0.255
  deny ip 10.0.76.0 0.0.3.255 10.0.91.56 0.0.0.3
  deny ip 10.0.76.0 0.0.3.255 10.0.91.52 0.0.0.3
  deny ip 10.0.76.0 0.0.3.255 10.0.91.60 0.0.0.3
  permit ip any any
ip access-list extended To_Network90
  permit udp 10.0.80.0 0.0.3.255 host 10.0.91.58 eq 123
  permit udp 10.0.80.0 0.0.3.255 host 10.0.91.54 eq bootps
  permit udp 10.0.80.0 0.0.3.255 host 10.0.91.54 eq bootpc
  permit udp 10.0.80.0 0.0.3.255 host 10.0.91.62 eq domain
  permit tcp 10.0.80.0 0.0.3.255 host 10.0.91.62 eq domain
  deny ip 10.0.80.0 0.0.3.255 172.16.1.0 0.0.0.255
  deny ip 10.0.80.0 0.0.3.255 10.0.91.56 0.0.0.3
  deny ip 10.0.80.0 0.0.3.255 10.0.91.52 0.0.0.3
  deny ip 10.0.80.0 0.0.3.255 10.0.91.60 0.0.0.3
  permit ip any any
access-list 1 permit host 172.16.1.129
!
line con 0
!
line aux 0
!
line vty 0 4
  access-class 1 in
  login local
  transport input ssh
line vty 5 15
  access-class 1 in
  login local
  transport input ssh
!
ntp server 10.0.91.58

```

```
!  
end
```

Multilyer-Switch-C-2:

```
!  
hostname Multilayer-Switch-C-2  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip routing  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip domain-name DavidsLab  
!  
spanning-tree mode pvst  
spanning-tree vlan 70 priority 24576  
spanning-tree vlan 80 priority 28672  
!  
interface Loopback0  
ip address 1.1.1.12 255.255.255.255  
!  
interface Port-channel1  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface Port-channel4  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface Port-channel6  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90,400  
switchport trunk encapsulation dot1q  
switchport mode trunk  
!  
interface FastEthernet0/1  
no switchport  
ip address 10.0.91.110 255.255.255.252  
duplex auto  
speed auto  
!  
interface FastEthernet0/2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 70,80,90,400  
switchport trunk encapsulation dot1q
```

```
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/3
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/4
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/7
no switchport
ip address 10.0.91.61 255.255.255.252
duplex auto
speed auto
!
interface FastEthernet0/8
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 4 mode desirable
!
interface FastEthernet0/9
switchport trunk native vlan 330
switchport trunk allowed vlan 70,80,90,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 4 mode desirable
!
interface Vlan1
no ip address
shutdown
!
interface Vlan7
mac-address 0090.0c07.1b01
no ip address
!
interface Vlan70
mac-address 0090.0c07.1b02
ip address 10.0.72.2 255.255.252.0
ip helper-address 10.0.91.54
ip access-group To_Network70 in
standby version 2
standby 7 ip 10.0.72.1
standby 0 preempt
!
interface Vlan80
mac-address 0090.0c07.1b03
```

```

ip address 10.0.76.2 255.255.252.0
ip helper-address 10.0.91.54
ip access-group To_Network80 in
standby version 2
standby 8 ip 10.0.76.1
standby 0 preempt
!
interface Vlan90
mac-address 0090.0c07.1b04
ip address 10.0.80.2 255.255.252.0
ip helper-address 10.0.91.54
ip access-group To_Network90 in
standby version 2
standby 9 ip 10.0.80.1
standby 0 preempt
!
interface Vlan400
mac-address 0090.0c07.1b05
ip address 172.16.1.93 255.255.255.224
standby version 2
standby 400 ip 172.16.1.94
!
router ospf 1
log-adjacency-changes
network 10.0.91.108 0.0.0.3 area 0
network 10.0.72.0 0.0.3.255 area 0
network 10.0.76.0 0.0.3.255 area 0
network 10.0.80.0 0.0.3.255 area 0
network 10.0.91.60 0.0.0.3 area 0
network 172.16.1.64 0.0.0.3 area 0
network 1.1.1.12 0.0.0.0 area 0
!
ip classless
!
ip flow-export version 9
!
!
ip access-list extended To_Network70
permit udp 10.0.72.0 0.0.3.255 host 10.0.91.58 eq 123
permit udp 10.0.72.0 0.0.3.255 host 10.0.91.54 eq bootps
permit udp 10.0.72.0 0.0.3.255 host 10.0.91.54 eq bootpc
permit udp 10.0.72.0 0.0.3.255 host 10.0.91.62 eq domain
permit tcp 10.0.72.0 0.0.3.255 host 10.0.91.62 eq domain
deny ip 10.0.72.0 0.0.3.255 172.16.1.0 0.0.0.255
deny ip 10.0.72.0 0.0.3.255 10.0.91.56 0.0.0.3
deny ip 10.0.72.0 0.0.3.255 10.0.91.52 0.0.0.3
deny ip 10.0.72.0 0.0.3.255 10.0.91.60 0.0.0.3
permit ip any any
ip access-list extended To_Network80
permit udp 10.0.76.0 0.0.3.255 host 10.0.91.58 eq 123
permit udp 10.0.76.0 0.0.3.255 host 10.0.91.54 eq bootps
permit udp 10.0.76.0 0.0.3.255 host 10.0.91.54 eq bootpc
permit udp 10.0.76.0 0.0.3.255 host 10.0.91.62 eq domain
permit tcp 10.0.76.0 0.0.3.255 host 10.0.91.62 eq domain
deny ip 10.0.76.0 0.0.3.255 172.16.1.0 0.0.0.255
deny ip 10.0.76.0 0.0.3.255 10.0.91.56 0.0.0.3
deny ip 10.0.76.0 0.0.3.255 10.0.91.52 0.0.0.3
deny ip 10.0.76.0 0.0.3.255 10.0.91.60 0.0.0.3
permit ip any any
ip access-list extended To_Network90
permit udp 10.0.80.0 0.0.3.255 host 10.0.91.58 eq 123
permit udp 10.0.80.0 0.0.3.255 host 10.0.91.54 eq bootps
permit udp 10.0.80.0 0.0.3.255 host 10.0.91.54 eq bootpc
permit udp 10.0.80.0 0.0.3.255 host 10.0.91.62 eq domain
permit tcp 10.0.80.0 0.0.3.255 host 10.0.91.62 eq domain
deny ip 10.0.80.0 0.0.3.255 172.16.1.0 0.0.0.255
deny ip 10.0.80.0 0.0.3.255 10.0.91.56 0.0.0.3
deny ip 10.0.80.0 0.0.3.255 10.0.91.52 0.0.0.3
deny ip 10.0.80.0 0.0.3.255 10.0.91.60 0.0.0.3
permit ip any any

```

```
access-list 1 permit host 172.16.1.129
!
line con 0
!
line aux 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
ntp server 10.0.91.58
!
end
```

Switch-D-1:

```
!
hostname Switch-D-1
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip domain-name DavidsLab
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
spanning-tree mode pvst
spanning-tree portfast default
spanning-tree portfast bpduguard default
spanning-tree extend system-id
spanning-tree vlan 120 priority 24576
spanning-tree vlan 400 priority 28672
!
interface Port-channel1
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport mode trunk
!
interface Port-channel2
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport mode trunk
!
interface Port-channel3
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport mode trunk
!
interface FastEthernet0/1
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/3
```

```

switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport mode trunk
channel-group 3 mode desirable
!
interface FastEthernet0/4
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/7
switchport access vlan 110
switchport mode access
switchport voice vlan 100
!
interface FastEthernet0/8
switchport access vlan 120
switchport mode access
!
interface FastEthernet0/9
switchport access vlan 120
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
interface Vlan400
ip address 172.16.1.97 255.255.255.224
!
ip default-gateway 172.16.1.126
!
access-list 1 permit host 172.16.1.129
line con 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
end

```

Switch-D-2:

```

!
hostname Switch-D-2
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip domain-name DavidsLab
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/

```



```
!  
spanning-tree mode pvst  
spanning-tree portfast default  
spanning-tree portfast bpduguard default  
spanning-tree extend system-id  
spanning-tree vlan 400 priority 24576  
spanning-tree vlan 120 priority 28672  
!  
interface Port-channel3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,400  
switchport mode trunk  
!  
interface Port-channel4  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,400  
switchport mode trunk  
!  
interface Port-channel5  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,400  
switchport mode trunk  
!  
interface FastEthernet0/1  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,400  
switchport mode trunk  
channel-group 5 mode desirable  
!  
interface FastEthernet0/2  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,400  
switchport mode trunk  
channel-group 3 mode desirable  
!  
interface FastEthernet0/3  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,400  
switchport mode trunk  
channel-group 3 mode desirable  
!  
interface FastEthernet0/4  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,400  
switchport mode trunk  
channel-group 5 mode desirable  
!  
interface FastEthernet0/5  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,400  
switchport mode trunk  
channel-group 4 mode desirable  
!  
interface FastEthernet0/6  
switchport trunk native vlan 330  
switchport trunk allowed vlan 100,110,120,400  
switchport mode trunk  
channel-group 4 mode desirable  
!  
interface FastEthernet0/7  
switchport access vlan 110  
switchport mode access  
switchport voice vlan 100  
!  
interface FastEthernet0/8  
switchport access vlan 120  
switchport mode access  
!  
interface Vlan1  
no ip address
```

```

shutdown
!
interface Vlan400
ip address 172.16.1.98 255.255.255.224
!
ip default-gateway 172.16.1.126
!
access-list 1 permit host 172.16.1.129
line con 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
end

```

Multilayer-Switch-D-1:

```

!
hostname Multilayer-Switch-D-1
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip routing
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip domain-name DavidsLab
!
spanning-tree mode pvst
spanning-tree vlan 110 priority 20480
spanning-tree vlan 100 priority 28672
!
interface Loopback0
ip address 1.1.1.17 255.255.255.255
!
interface Port-channel1
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Port-channel4
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Port-channel6
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface FastEthernet0/1
no switchport
ip address 10.0.91.122 255.255.255.252
duplex auto

```

```
speed auto
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/3
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/4
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 1 mode desirable
!
interface FastEthernet0/7
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 4 mode desirable
!
interface FastEthernet0/8
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 4 mode desirable
!
interface Vlan1
no ip address
shutdown
!
interface Vlan100
mac-address 0060.3e76.7701
ip address 10.0.84.2 255.255.254.0
ip helper-address 10.0.91.54
ip access-group To_Network100 in
standby version 2
standby 100 ip 10.0.84.1
standby 0 preempt
!
interface Vlan110
mac-address 0060.3e76.7702
ip address 10.0.86.2 255.255.254.0
ip helper-address 10.0.91.54
ip access-group To_Network110 in
standby version 2
```

```

standby 110 ip 10.0.86.1
standby 0 preempt
!
interface Vlan120
mac-address 0060.3e76.7703
ip address 10.0.88.2 255.255.254.0
ip helper-address 10.0.91.54
ip access-group To_Network120 in
standby version 2
standby 120 ip 10.0.88.1
standby 0 preempt
!
interface Vlan400
mac-address 0060.3e76.7704
ip address 172.16.1.124 255.255.255.224
standby version 2
standby 400 ip 172.16.1.126
!
router ospf 1
log-adjacency-changes
network 10.0.91.120 0.0.0.3 area 0
network 10.0.84.0 0.0.1.255 area 0
network 10.0.86.0 0.0.1.255 area 0
network 10.0.88.0 0.0.1.255 area 0
network 172.16.1.96 0.0.0.31 area 0
network 1.1.1.17 0.0.0.0 area 0
!
ip classless
!
ip flow-export version 9
!
ip access-list extended To_Network100
permit udp 10.0.84.0 0.0.1.255 host 10.0.91.58 eq 123
permit udp 10.0.84.0 0.0.1.255 host 10.0.91.54 eq bootps
permit udp 10.0.84.0 0.0.1.255 host 10.0.91.54 eq bootpc
permit udp 10.0.84.0 0.0.1.255 host 10.0.91.62 eq domain
permit tcp 10.0.84.0 0.0.1.255 host 10.0.91.62 eq domain
deny ip 10.0.84.0 0.0.1.255 172.16.1.0 0.0.0.255
deny ip 10.0.84.0 0.0.1.255 10.0.91.56 0.0.0.3
deny ip 10.0.84.0 0.0.1.255 10.0.91.52 0.0.0.3
deny ip 10.0.84.0 0.0.1.255 10.0.91.60 0.0.0.3
permit ip any any
ip access-list extended To_Network110
permit udp 10.0.86.0 0.0.1.255 host 10.0.91.58 eq 123
permit udp 10.0.86.0 0.0.1.255 host 10.0.91.54 eq bootps
permit udp 10.0.86.0 0.0.1.255 host 10.0.91.54 eq bootpc
permit udp 10.0.86.0 0.0.1.255 host 10.0.91.62 eq domain
permit tcp 10.0.86.0 0.0.1.255 host 10.0.91.62 eq domain
deny ip 10.0.86.0 0.0.1.255 172.16.1.0 0.0.0.255
deny ip 10.0.86.0 0.0.1.255 10.0.91.56 0.0.0.3
deny ip 10.0.86.0 0.0.1.255 10.0.91.52 0.0.0.3
deny ip 10.0.86.0 0.0.1.255 10.0.91.60 0.0.0.3
permit ip any any
ip access-list extended To_Network120
permit udp 10.0.88.0 0.0.1.255 host 10.0.91.58 eq 123
permit udp 10.0.88.0 0.0.1.255 host 10.0.91.54 eq bootps
permit udp 10.0.88.0 0.0.1.255 host 10.0.91.54 eq bootpc
permit udp 10.0.88.0 0.0.1.255 host 10.0.91.62 eq domain
permit tcp 10.0.88.0 0.0.1.255 host 10.0.91.62 eq domain
deny ip 10.0.88.0 0.0.1.255 172.16.1.0 0.0.0.255
deny ip 10.0.88.0 0.0.1.255 10.0.91.56 0.0.0.3
deny ip 10.0.88.0 0.0.1.255 10.0.91.52 0.0.0.3
deny ip 10.0.88.0 0.0.1.255 10.0.91.60 0.0.0.3
permit ip any any
access-list 1 permit host 172.16.1.129
!
line con 0
!
line aux 0
!

```

```
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
ntp server 10.0.91.58
!
end
```

Multilayer-Switch-D-2:

```
!
hostname Multilayer-Switch-D-2
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip routing
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip domain-name DavidsLab
!
spanning-tree mode pvst
spanning-tree vlan 100 priority 20480
spanning-tree vlan 110 priority 28672
!
interface Loopback0
ip address 1.1.1.16 255.255.255.255
!
interface Port-channel2
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Port-channel5
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface Port-channel6
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
!
interface FastEthernet0/1
no switchport
```

```
ip address 10.0.91.126 255.255.255.252
duplex auto
speed auto
!
interface FastEthernet0/2
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 5 mode desirable
!
interface FastEthernet0/3
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/4
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/5
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 6 mode desirable
!
interface FastEthernet0/6
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/7
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 2 mode desirable
!
interface FastEthernet0/8
switchport trunk native vlan 330
switchport trunk allowed vlan 100,110,120,400
switchport trunk encapsulation dot1q
switchport mode trunk
channel-group 5 mode desirable
!
interface FastEthernet0/9
switchport mode access
!
interface Vlan1
no ip address
shutdown
!
interface Vlan100
mac-address 000c.857a.0c01
ip address 10.0.84.3 255.255.254.0
ip helper-address 10.0.91.54
ip access-group To_Network100 in
standby version 2
standby 100 ip 10.0.84.1
!
interface Vlan110
mac-address 000c.857a.0c02
```

```

ip address 10.0.86.3 255.255.254.0
ip helper-address 10.0.91.54
ip access-group To_Network110 in
standby version 2
standby 110 ip 10.0.86.1
!
interface Vlan120
mac-address 000c.857a.0c03
ip address 10.0.88.3 255.255.254.0
ip helper-address 10.0.91.54
ip access-group To_Network120 in
standby version 2
standby 120 ip 10.0.88.1
!
interface Vlan400
mac-address 000c.857a.0c04
ip address 172.16.1.125 255.255.255.224
standby version 2
standby 0 preempt
standby 400 ip 172.16.1.126
!
router ospf 1
log-adjacency-changes
network 10.0.91.124 0.0.0.3 area 0
network 10.0.84.0 0.0.1.255 area 0
network 10.0.86.0 0.0.1.255 area 0
network 10.0.88.0 0.0.1.255 area 0
network 172.16.1.96 0.0.0.31 area 0
network 1.1.1.16 0.0.0.0 area 0
!
ip classless
!
ip flow-export version 9
!
ip access-list extended To_Network100
permit udp 10.0.84.0 0.0.1.255 host 10.0.91.58 eq 123
permit udp 10.0.84.0 0.0.1.255 host 10.0.91.54 eq bootps
permit udp 10.0.84.0 0.0.1.255 host 10.0.91.54 eq bootpc
permit udp 10.0.84.0 0.0.1.255 host 10.0.91.62 eq domain
permit tcp 10.0.84.0 0.0.1.255 host 10.0.91.62 eq domain
deny ip 10.0.84.0 0.0.1.255 172.16.1.0 0.0.0.255
deny ip 10.0.84.0 0.0.1.255 10.0.91.56 0.0.0.3
deny ip 10.0.84.0 0.0.1.255 10.0.91.52 0.0.0.3
deny ip 10.0.84.0 0.0.1.255 10.0.91.60 0.0.0.3
permit ip any any
ip access-list extended To_Network110
permit udp 10.0.86.0 0.0.1.255 host 10.0.91.58 eq 123
permit udp 10.0.86.0 0.0.1.255 host 10.0.91.54 eq bootps
permit udp 10.0.86.0 0.0.1.255 host 10.0.91.54 eq bootpc
permit udp 10.0.86.0 0.0.1.255 host 10.0.91.62 eq domain
permit tcp 10.0.86.0 0.0.1.255 host 10.0.91.62 eq domain
deny ip 10.0.86.0 0.0.1.255 172.16.1.0 0.0.0.255
deny ip 10.0.86.0 0.0.1.255 10.0.91.56 0.0.0.3
deny ip 10.0.86.0 0.0.1.255 10.0.91.52 0.0.0.3
deny ip 10.0.86.0 0.0.1.255 10.0.91.60 0.0.0.3
permit ip any any
ip access-list extended To_Network120
permit udp 10.0.88.0 0.0.1.255 host 10.0.91.58 eq 123
permit udp 10.0.88.0 0.0.1.255 host 10.0.91.54 eq bootps
permit udp 10.0.88.0 0.0.1.255 host 10.0.91.54 eq bootpc
permit udp 10.0.88.0 0.0.1.255 host 10.0.91.62 eq domain
permit tcp 10.0.88.0 0.0.1.255 host 10.0.91.62 eq domain
deny ip 10.0.88.0 0.0.1.255 172.16.1.0 0.0.0.255
deny ip 10.0.88.0 0.0.1.255 10.0.91.56 0.0.0.3
deny ip 10.0.88.0 0.0.1.255 10.0.91.52 0.0.0.3
deny ip 10.0.88.0 0.0.1.255 10.0.91.60 0.0.0.3
permit ip any any
access-list 1 permit host 172.16.1.129
!
line con 0

```

```

!
line aux 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
ntp server 10.0.91.58
!
end

```

Multilayer-Switch-E:

```

!
hostname Multilayer-Switch-E
!
ip routing
!
spanning-tree mode pvst
!
interface FastEthernet0/1
no switchport
ip address 10.0.91.130 255.255.255.252
duplex auto
speed auto
!
interface FastEthernet0/2
switchport access vlan 400
switchport mode access

interface Vlan1
no ip address
shutdown
!
interface Vlan400
mac-address 0001.4372.b401
ip address 172.16.1.158 255.255.255.224
!
router ospf 1
log-adjacency-changes
network 10.0.91.128 0.0.0.3 area 0
network 172.16.1.128 0.0.0.31 area 0
!
ip classless
!
ip flow-export version 9
!
line con 0
!
line aux 0
!
line vty 0 4
login
!
ntp server 10.0.91.58
!
end

```


Router-A-1:

```
!  
hostname Router-A-1  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip cef  
no ipv6 cef  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
license udi pid CISCO2911/K9 sn FTX15241WQ1-  
!  
ip domain-name Davidslab  
!  
spanning-tree mode pvst  
!  
interface Loopback0  
ip address 1.1.1.5 255.255.255.255  
!  
interface GigabitEthernet0/0  
ip address 10.0.91.33 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
ip address 10.0.91.74 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/2  
ip address 10.0.91.78 255.255.255.252  
duplex auto  
speed auto  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router ospf 1  
log-adjacency-changes  
network 10.0.91.32 0.0.0.3 area 0  
network 10.0.91.72 0.0.0.3 area 0  
network 10.0.91.76 0.0.0.3 area 0  
network 1.1.1.5 0.0.0.0 area 0  
!  
ip classless  
!  
ip flow-export version 9  
!  
access-list 1 permit host 172.16.1.129  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
access-class 1 in  
login local  
transport input ssh  
line vty 5 15  
access-class 1 in  
login local  
transport input ssh  
!  
ntp server 10.0.91.58  
!  
end
```

Router-A-2:

```
!  
hostname Router-A-2  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip cef  
no ipv6 cef  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
license udi pid CISCO2911/K9 sn FTX15242ZL9-  
!  
ip domain-name Davidslab  
!  
spanning-tree mode pvst  
!  
interface Loopback0  
ip address 1.1.1.3 255.255.255.255  
!  
interface GigabitEthernet0/0  
ip address 10.0.91.73 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
ip address 10.0.91.66 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/2  
no ip address  
duplex auto  
speed auto  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router ospf 1  
log-adjacency-changes  
network 10.0.91.72 0.0.0.3 area 0  
network 10.0.91.64 0.0.0.3 area 0  
network 1.1.1.3 0.0.0.0 area 0  
!  
ip classless  
!  
ip flow-export version 9  
!  
access-list 1 permit host 172.16.1.129  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
login local  
transport input ssh  
line vty 5 15  
login local  
transport input ssh  
!  
ntp server 10.0.91.58  
!  
end
```

Router-A-3:

```
!  
hostname Router-A-3  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip cef  
no ipv6 cef  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
license udi pid CISCO2911/K9 sn FTX15241KY1-  
!  
ip domain-name DavidsLab  
!  
spanning-tree mode pvst  
!  
interface Loopback0  
ip address 1.1.1.4 255.255.255.255  
!  
interface GigabitEthernet0/0  
ip address 10.0.91.77 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
ip address 10.0.91.70 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/2  
no ip address  
duplex auto  
speed auto  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router ospf 1  
log-adjacency-changes  
network 10.0.91.76 0.0.0.3 area 0  
network 10.0.91.68 0.0.0.3 area 0  
network 1.1.1.4 0.0.0.0 area 0  
!  
ip classless  
!  
ip flow-export version 9  
!  
access-list 1 permit host 172.16.1.129  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
access-class 1 in  
login local  
transport input ssh  
line vty 5 15  
access-class 1 in  
login local  
transport input ssh  
!  
ntp server 10.0.91.58  
!  
end
```

Router-B-1:

```
!  
hostname Router-B-1  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip cef  
no ipv6 cef  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
license udi pid CISCO2911/K9 sn FTX15240XI2-  
!  
ip domain-name DavidsLab  
!  
spanning-tree mode pvst  
!  
interface Loopback0  
ip address 1.1.1.10 255.255.255.255  
!  
interface GigabitEthernet0/0  
ip address 10.0.91.37 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
ip address 10.0.91.86 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/2  
ip address 10.0.91.82 255.255.255.252  
duplex auto  
speed auto  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router ospf 1  
log-adjacency-changes  
network 10.0.91.36 0.0.0.3 area 0  
network 10.0.91.80 0.0.0.3 area 0  
network 10.0.91.84 0.0.0.3 area 0  
network 1.1.1.10 0.0.0.0 area 0  
!  
ip classless  
!  
ip flow-export version 9  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
access-class 1 in  
login local  
transport input ssh  
line vty 5 15  
access-class 1 in  
login local  
transport input ssh  
!  
ntp server 10.0.91.58  
!  
end
```

Router-B-2:

```
!  
hostname Router-B-1  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip cef  
no ipv6 cef  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
license udi pid CISCO2911/K9 sn FTX15243GEP-  
!  
ip domain-name DavidsLab  
!  
spanning-tree mode pvst  
!  
interface Loopback0  
ip address 1.1.1.9 255.255.255.255  
!  
interface GigabitEthernet0/0  
ip address 10.0.91.85 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
ip address 10.0.91.94 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/2  
no ip address  
duplex auto  
speed auto  
shutdown  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router ospf 1  
log-adjacency-changes  
network 10.0.91.84 0.0.0.3 area 0  
network 10.0.91.92 0.0.0.3 area 0  
network 1.1.1.9 0.0.0.0 area 0  
!  
ip classless  
!  
ip flow-export version 9  
!  
access-list 1 permit host 172.16.1.129  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
access-class 1 in  
login local  
transport input ssh  
line vty 5 15  
access-class 1 in  
login local  
transport input ssh  
!  
ntp server 10.0.91.58  
!  
end
```

Router-B-3:

```
!  
hostname Router-B-3  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip cef  
no ipv6 cef  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
license udi pid CISCO2911/K9 sn FTX1524YZ3P-  
!  
ip domain-name DavidsLab  
!  
spanning-tree mode pvst  
!  
interface Loopback0  
ip address 1.1.1.8 255.255.255.255  
!  
interface GigabitEthernet0/0  
ip address 10.0.91.81 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
ip address 10.0.91.90 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/2  
no ip address  
duplex auto  
speed auto  
shutdown  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router ospf 1  
log-adjacency-changes  
network 10.0.91.80 0.0.0.3 area 0  
network 10.0.91.88 0.0.0.3 area 0  
network 1.1.1.8 0.0.0.0 area 0  
!  
ip classless  
!  
ip flow-export version 9  
!  
access-list 1 permit host 172.16.1.129  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
access-class 1 in  
login local  
transport input ssh  
line vty 5 15  
access-class 1 in  
login local  
transport input ssh  
!  
ntp server 10.0.91.58  
!
```

end

Router-C-1:

```
!  
hostname Router-C-1  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip cef  
no ipv6 cef  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
license udi pid CISCO2911/K9 sn FTX152401X0-  
!  
ip domain-name DavidsLab  
!  
!  
spanning-tree mode pvst  
!  
interface Loopback0  
ip address 1.1.1.15 255.255.255.255  
!  
interface GigabitEthernet0/0  
ip address 10.0.91.41 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
ip address 10.0.91.97 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/2  
ip address 10.0.91.101 255.255.255.252  
duplex auto  
speed auto  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router ospf 1  
log-adjacency-changes  
network 10.0.91.40 0.0.0.3 area 0  
network 10.0.91.96 0.0.0.3 area 0  
network 10.0.91.100 0.0.0.3 area 0  
network 1.1.1.15 0.0.0.0 area 0  
!  
ip classless  
!  
ip flow-export version 9  
!  
access-list 1 permit host 172.16.1.129  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
access-class 1 in  
login local  
transport input ssh  
line vty 5 15  
access-class 1 in  
login local  
transport input ssh
```

```
!  
ntp server 10.0.91.58  
!  
end
```

Router-C-2:

```
!  
hostname Router-C-2  
!  
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
ip cef  
no ipv6 cef  
!  
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/  
!  
license udi pid CISCO2911/K9 sn FTX15240IMT-  
!  
ip domain-name DavidsLab  
!  
spanning-tree mode pvst  
!  
interface Loopback0  
ip address 1.1.1.13 255.255.255.255  
!  
interface GigabitEthernet0/0  
ip address 10.0.91.98 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/1  
ip address 10.0.91.105 255.255.255.252  
duplex auto  
speed auto  
!  
interface GigabitEthernet0/2  
no ip address  
duplex auto  
speed auto  
shutdown  
!  
interface Vlan1  
no ip address  
shutdown  
!  
router ospf 1  
log-adjacency-changes  
network 10.0.91.96 0.0.0.3 area 0  
network 10.0.91.104 0.0.0.3 area 0  
network 1.1.1.13 0.0.0.0 area 0  
!  
ip classless  
!  
ip flow-export version 9  
!  
access-list 1 permit host 172.16.1.129  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
access-class 1 in  
login local  
transport input ssh  
line vty 5 15  
access-class 1 in
```



```
login local
transport input ssh
!
ntp server 10.0.91.58
!
end
```

Router-C-3:

```
!
hostname Router-C-3
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip cef
no ipv6 cef
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
license udi pid CISCO2911/K9 sn FTX15242DGM-
!
ip domain-name DavidsLab
!
!
spanning-tree mode pvst
!
interface Loopback0
ip address 1.1.1.14 255.255.255.255
!
interface GigabitEthernet0/0
ip address 10.0.91.102 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 10.0.91.109 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 10.0.91.100 0.0.0.3 area 0
network 10.0.91.108 0.0.0.3 area 0
network 1.1.1.14 0.0.0.0 area 0
!
ip classless
!
ip flow-export version 9
!
access-list 1 permit host 172.16.1.129
!
line con 0
!
line aux 0
!
line vty 0 4
access-class 1 in
login local
```

```

transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
ntp server 10.0.91.58
!
end

```

Router-D-1:

```

!
hostname Router-D-1
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip cef
no ipv6 cef
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
license udi pid CISCO2911/K9 sn FTX1524MJ6C-
!
ip domain-name DavidsLab
!
spanning-tree mode pvst
!
interface Loopback0
ip address 1.1.1.20 255.255.255.255
!
interface GigabitEthernet0/0
ip address 10.0.91.45 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 10.0.91.117 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/2
ip address 10.0.91.113 255.255.255.252
duplex auto
speed auto
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 10.0.91.44 0.0.0.3 area 0
network 10.0.91.112 0.0.0.3 area 0
network 10.0.91.116 0.0.0.3 area 0
network 1.1.1.20 0.0.0.0 area 0
!
ip classless
!
ip flow-export version 9
!
access-list 1 permit host 172.16.1.129
!
line con 0
!
line aux 0
!

```

```
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
ntp server 10.0.91.58
!
end
```

Router-D-2:

```
!
hostname Router-D-2
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip cef
no ipv6 cef
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
license udi pid CISCO2911/K9 sn FTX15249PXR-
!
ip domain-name DavidsLab
!
spanning-tree mode pvst
!
interface Loopback0
ip address 1.1.1.18 255.255.255.255
!
interface GigabitEthernet0/0
ip address 10.0.91.114 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 10.0.91.121 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/2
no ip address
duplex auto
speed auto
shutdown
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 10.0.91.112 0.0.0.3 area 0
network 10.0.91.120 0.0.0.3 area 0
network 1.1.1.18 0.0.0.0 area 0
!
ip classless
!
ip flow-export version 9
!
access-list 1 permit host 172.16.1.129
!
line con 0
```

```

!
line aux 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
ntp server 10.0.91.58
!
end

```

Router-D-3:

```

!
hostname Router-D-3
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip cef
no ipv6 cef
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
license udi pid CISCO2911/K9 sn FTX1524OFYE-
!
ip domain-name DavidsLab
!
spanning-tree mode pvst
!
interface Loopback0
ip address 1.1.1.19 255.255.255.255
!
interface GigabitEthernet0/0
ip address 10.0.91.118 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 10.0.91.125 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/2
ip address 10.0.91.129 255.255.255.252
duplex auto
speed auto
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 10.0.91.116 0.0.0.3 area 0
network 10.0.91.124 0.0.0.3 area 0
network 10.0.91.128 0.0.0.3 area 0
network 1.1.1.19 0.0.0.0 area 0
!
ip classless
!
ip flow-export version 9
!
access-list 1 permit host 172.16.1.129

```

```

!
line con 0
!
line aux 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
ntp server 10.0.91.58
!
end

```

Router-E-1:

```

!
hostname Router-E-1
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip cef
no ipv6 cef
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
license udi pid CISCO2911/K9 sn FTX15249693-
!
ip domain-name DavidsLab
!
spanning-tree mode pvst
!
interface Loopback0
ip address 1.1.1.21 255.255.255.255
!
interface GigabitEthernet0/0
ip address 10.0.91.49 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 10.0.91.38 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/2
ip address 10.0.91.34 255.255.255.252
duplex auto
speed auto
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 10.0.91.32 0.0.0.3 area 0
network 10.0.91.36 0.0.0.3 area 0
network 10.0.91.48 0.0.0.3 area 0
network 1.1.1.21 0.0.0.0 area 0
!
ip classless
!
ip flow-export version 9

```

```

!
access-list 1 permit host 172.16.1.129
!
line con 0
!
line aux 0
!
line vty 0 4
access-class 1 in
login local
transport input ssh
line vty 5 15
access-class 1 in
login local
transport input ssh
!
ntp server 10.0.91.58
!
end

```

Router-E-2:

```

!
hostname Router-E-2
!
enable secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
ip cef
no ipv6 cef
!
username David secret 5 $1$mERr$CunnQjkAEasAf02ni13tK/
!
license udi pid CISCO2911/K9 sn FTX1524SLH2-
!
ip domain-name DavidsLab
!
spanning-tree mode pvst
!
interface Loopback0
ip address 1.1.1.22 255.255.255.255
!
interface GigabitEthernet0/0
ip address 10.0.91.50 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/1
ip address 10.0.91.42 255.255.255.252
duplex auto
speed auto
!
interface GigabitEthernet0/2
ip address 10.0.91.46 255.255.255.252
duplex auto
speed auto
!
interface Vlan1
no ip address
shutdown
!
router ospf 1
log-adjacency-changes
network 10.0.91.44 0.0.0.3 area 0
network 10.0.91.40 0.0.0.3 area 0
network 10.0.91.48 0.0.0.3 area 0
network 1.1.1.22 0.0.0.0 area 0
!
ip classless

```

```
!  
ip flow-export version 9  
!  
access-list 1 permit host 172.16.1.129  
!  
line con 0  
!  
line aux 0  
!  
line vty 0 4  
access-class 1 in  
login local  
transport input ssh  
line vty 5 15  
access-class 1 in  
login local  
transport input ssh  
!  
ntp server 10.0.91.58  
!  
end
```